

**Amrita School of Engineering, Bengaluru**  
**Department of ECE**  
**B. Tech. AY 2025-2026 (VII Semester)**  
**Synopsis – 23ECE498 & 23EAC498/Project Phase-1**

**Batch Number/Batch ID:** B008

**Name of the Students and Signature :**

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**Register number of the students:**

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**Name of Guide and Signature:** Dr. Sreeja Kochuvila

**Name of the Domain or TAG:** Robotics, Embodied AI

**Title of the Project Phase-1:** Framework for Autonomous Drone Operation Using LLMs & VLMs

**Project Phase-1 Objectives along with the sub-objectives:**

Primary Objective: To develop a modular, closed-loop drone framework for navigation and task execution that translates high-level natural language commands into autonomous actions using LLMs for task planning, VLMs for overhead camera-based perception, and an ESP32-S3 drone for flight execution.

Sub-objectives:

1. Design modular software architecture separating perception (VLM), planning (LLM), and low-level control.
2. Develop a coordinate transformation system to map overhead camera pixel inputs into real-world drone waypoints.
3. Implement low-level drone skills (takeoff, hover, translation, landing) on ESP32-S3 firmware.
4. Build a feedback loop allowing the drone to replan and adapt dynamically when objects move or flight errors occur.

5. Benchmark the framework against static/non-adaptive flight sequences, targeting task-success rates comparable to or exceeding recent simulation-based results (e.g. ~79% in Gamage et al. [4]).

### References:

- [1] Y. Emami, H. Zhou, R. Reddy, A. H. Arani, B. Wang, K. Li, L. Almeida, and Z. Han, "Large language model-assisted UAV operations and communications: A multifaceted survey and tutorial," *arXiv preprint arXiv:2602.19534*, 2026.
- [2] H. Chen, J. Zheng, S. Yang, T. Zeng, S. Feng, S. Cheng, R. Ren, H. Guo, S. Yuan, X. Wang, K. Wang, and J. Pei, "Vision-and-language navigation for UAVs: Progress, challenges, and a research roadmap," *arXiv preprint arXiv:2604.13654*, 2026.
- [3] M. Krupáš, Ľ. Urbík, and I. Zolotová, "Multimodal AI for UAV: Vision–language models in human–machine collaboration," *Electronics*, vol. 14, no. 17, p. 3548, 2025.
- [4] L. Yuan, C. Deng, D.-J. Han, I. Hwang, S. Brunswicker, and C. G. Brinton, "Next-generation LLM for UAV: From natural language to autonomous flight," *arXiv preprint arXiv:2510.21739*, 2025.
- [5] D. Driess, F. Xia, M. S. M. Sajjadi, C. Lynch, A. Chowdhery, B. Ichter, A. Wahid, J. Tompson, Q. Vuong, T. Yu, W. Huang, Y. Chebotar, P. Sermanet, D. Duckworth, S. Levine, V. Vanhoucke, K. Hausman, M. Toussaint, K. Greff, A. Zeng, I. Mordatch, and P. Florence, "PaLM-E: An embodied multimodal language model," in *Proc. Int. Conf. Machine Learning (ICML)*, 2023, pp. 8469–8488.
- [6] G. Manfredi, U. Erra, N. Capece, and M. Rinaldi, "FlyAI: A unified NLP and computer vision framework for autonomous drone mission programming," *CEUR Workshop Proceedings*, vol. 4103, paper 3, 2025.
- [7] G. Chen, X. Yu, N. Ling, and L. Zhong, "TypeFly: Flying drones with large language model," *arXiv preprint arXiv:2312.14950*, 2023.

### Time Plan of Project Phase-1:

| Month          | Work to be implemented                                                                                                 |
|----------------|------------------------------------------------------------------------------------------------------------------------|
| July 2026      | Literature Review, Project Scoping, and Synopsis Formulation (Completed).                                              |
| August 2026    | Hardware Procurement, Component Sourcing, and Setup of Test Environment.                                               |
| September 2026 | Development of Perception Module (VLM) & Spatial Object Grounding.                                                     |
| October 2026   | LLM Command Parser & Mission Planner Integration.                                                                      |
| November 2026  | Drone Firmware Integration, Simulation Testing, Real-World Flight Trials, Documentation, and Final Report Preparation. |